AN AFFINE ETALE NEIGHBOURHOOD ON WHICH THE GLOBAL SECTIONS OF A FLAT FINITELY PRESENTED MODULE ARE PROJECTIVE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Openness of flatness. More on Morphisms, label more-morphisms-theorem-openness-flatness, invoked at proof line 5677. Lets the module be assumed flat over the whole of the shrunk source, which is the first reduction of the proof.
- The free and projective variants of the flat along the fibre lemma, and purity of an affine locally projective module. Same chapter flat.tex, labels lemma-finite-presentation-flat-along-fibre (proof line 5684) and lemma-affine-locally-projective-pure (proof line 5696). Produces the affine etale neighbourhood and the etale g with X_s inside g(X') and projective global sections, and gives universal purity of the pulled back module.
- Purity of a flat finitely presented module on an open, the projective criterion, and specialization of associated points along an etale morphism of fibres. Same chapter flat.tex, labels lemma-flat-finite-presentation-purity-open (proof line 5703), proposition-finite-presentation-flat-pure-is-projective (proof line 5705), lemma-associated-point-specializes (proof line 5712) and lemma-etale-weak-assassin-up-down (proof line 5716). The first two upgrade purity of the restriction to projectivity of its global sections, and the others supply the specialization of an associated point of the restriction into the fibre, obtained by lifting the point along the etale morphism and using purity of the pulled back module.

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Source reference. Original lemma 05K6: sources/stacks-flat/pinned-original-source.tex, source lines 5148–5148. Navigation pointer for \ref{lemma-affine-locally-projective-pure} used inside the counted span (line 5696); the argument that consumes it is retained in the span.

Source reference. Original lemma 05L3: sources/stacks-flat/pinned-original-source.tex, source lines 5175–5175. Navigation pointer for \ref{lemma-associated-point-specializes} used inside the counted span (line 5712); the argument that consumes it is retained in the span.

Source reference. Original lemma 05FP: sources/stacks-flat/pinned-original-source.tex, source lines 353–353. Navigation pointer for \ref{lemma-etale-weak-assassin-up-down} used inside the counted span (line 5716); the argument that consumes it is retained in the span.

Source reference. Original lemma 05KW: sources/stacks-flat/pinned-original-source.tex, source lines 3342–3342. Navigation pointer for \ref{lemma-finite-presentation-flat-along-fibre} used inside the counted span (line 5684); the argument that consumes it is retained in the span.

Source reference. Original lemma 05MC: sources/stacks-flat/pinned-original-source.tex, source lines 5490–5490. Navigation pointer for \ref{lemma-flat-finite-presentation-purity-open} used inside the counted span (line 5703); the argument that consumes it is retained in the span.

Source reference. Original lemma 0399: sources/stacks-chow/tags.txt, source lines 4264–4264. Navigation pointer for \ref{more-morphisms-theorem-openness-flatness} used inside the counted span (line 5677); the argument that consumes it is retained in the span.

Source reference. Original lemma 05MD: sources/stacks-flat/pinned-original-source.tex, source lines 5582–5582. Navigation pointer for \ref{proposition-finite-presentation-flat-pure-is-projective} used inside the counted span (line 5705); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to S$ be a morphism which is locally of finite presentation. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module which is of finite presentation. Let $x \in X$ with $s = f(x) \in S$. If $\mathcal{F}$ is flat at x over S there exists an affine elementary étale neighbourhood $(S', s') \to (S, s)$ and an affine open $U' \subset X \times_S S'$ which contains $x' = (x, s')$ such that $\Gamma(U', \mathcal{F}|_{U'})$ is a projective $\Gamma(S', \mathcal{O}_{S'})$-module.*

Proof. During the proof we may replace X by an open neighbourhood of x and we may replace S by an elementary étale neighbourhood of s. Hence, by openness of flatness (see More on Morphisms, Theorem 0399) we may assume that $\mathcal{F}$ is flat over S. We may assume S and X are affine. After shrinking X some more we may assume that any point of $\mathrm{Ass}_{X_s}(\mathcal{F}_s)$ is a generalization of x. This property is preserved on replacing (S, s) by an elementary étale neighbourhood. Hence we may apply Lemma 05KW to arrive at the situation where there exists a diagram

$$\begin{array}{ccc} X & \xleftarrow{\quad g \quad} & X' \\ & \searrow \quad \swarrow & \\ & S & \end{array}$$

of schemes affine and of finite presentation over S, where g is étale, $X_s \subset g(X')$, and with $\Gamma(X', g^*\mathcal{F})$ a projective $\Gamma(S, \mathcal{O}_S)$-module. Note that in this case $g^*\mathcal{F}$ is universally pure over S, see Lemma 05K6.

Let $U \subset g(X')$ be an affine open neighbourhood of x. We claim that $\mathcal{F}|_U$ is pure along U_s. If we prove this, then the lemma follows because $\mathcal{F}|_U$ will be pure relative to S after shrinking S, see Lemma 05MC, whereupon the projectivity follows from Proposition 05MD. To prove the claim we have to show, after replacing (S, s) by an arbitrary elementary étale neighbourhood, that any point ξ of $\mathrm{Ass}_{U/S}(\mathcal{F}|_U)$ lying over some $s' \in S$, $s' \rightsquigarrow s$ specializes to a point of U_s. Since $U \subset g(X')$ we can find a $\xi' \in X'$ with $g(\xi') = \xi$. Because $g^*\mathcal{F}$ is pure over S, using Lemma 05L3, we see there exists a specialization $\xi' \rightsquigarrow x'$ with $x' \in \mathrm{Ass}_{X'_s}(g^*\mathcal{F}_s)$. Then $g(x') \in \mathrm{Ass}_{X_s}(\mathcal{F}_s)$ (see for example Lemma 05FP applied to the étale morphism $X'_s \to X_s$ of Noetherian schemes) and hence $g(x') \rightsquigarrow x$ by our choice of X above! Since $x \in U$ we conclude that $g(x') \in U$. Thus $\xi = g(\xi') \rightsquigarrow g(x') \in U_s$ as desired. $\square$